

Complete SDET Setup Guide — Day 1

Goal

Build a professional SDET environment for:

- JavaScript
- Playwright
- Appium
- API Testing
- Git/GitHub workflow

This guide explains:

- what we installed
 - why we installed it
 - commands used
 - purpose behind each command
 - real-world examples
 - issues faced and fixes
-

1. VS Code Setup

What is VS Code?

VS Code is a code editor used to:

- write code
- run terminal commands
- debug applications
- manage GitHub projects

Why We Need It

Real SDETs write:

- automation scripts
- API tests
- framework code
- debugging utilities

VS Code becomes the main workspace.

Real-World Example

A Playwright automation engineer writes:

```
await page.click('#login');
```

inside VS Code.

2. Node.js Setup

Command Used

```
node -v
```

Purpose

Checks whether Node.js is installed.

Output

```
v22.13.1
```

What is Node.js?

Node.js allows JavaScript to run outside the browser.

Why Important for SDET?

Playwright, Appium JS, API automation all run using Node.js.

Without Node.js:

- JavaScript automation cannot run
- npm packages cannot install

Real-World Example

When running:

```
npx playwright test
```

Node.js executes the Playwright automation.

3. npm Setup

Command Used

```
npm -v
```

What is npm?

npm = Node Package Manager.

It installs libraries/packages.

Why Important?

Automation frameworks depend on npm packages.

Examples:

- Playwright
- Appium
- Axios
- Jest

Real-World Example

Installing Playwright:

```
npm init playwright@latest
```

Installing Appium:

```
npm install -g appium
```

4. Git Setup

Command Used

```
git --version
```

What is Git?

Git is a version control system.

Why Important?

Tracks:

- code changes
- bug fixes
- framework updates
- team collaboration

Real-World Example

Suppose framework breaks after new code. Git allows reverting back safely.

5. GitHub Repository Setup

Repository Name

```
sdet-journey_2026may
```

Purpose

Stores project online.

Why Important?

Acts as:

- portfolio
- backup

- interview proof
 - collaboration platform
-

6. Folder Structure

Created Structure

```
SDET-Journey
|
├─ javascript-practice
├─ coding-problems
├─ playwright-learning
├─ appium-learning
├─ api-testing
└─ notes
```

Why This Structure?

Keeps learning organized.

Real-World Example

Large automation projects contain:

- framework folder
- utilities
- tests
- reports
- API validations

Organization matters.

7. Git Initialization

Command Used

```
git init
```

Purpose

Converts folder into Git repository.

What Happens Internally?

Git creates hidden:

```
.git
```

folder.

This tracks all future code changes.

Real-World Example

Every automation framework in companies uses Git.

8. Connect GitHub Repository

Command Used

```
git remote add origin https://github.com/AshrafiFirdos/sdet-journey_2026may.git
```

Purpose

Connects local project ↔ GitHub repository.

What Does "origin" Mean?

"origin" is nickname for GitHub repo.

Real-World Example

Now local code can be pushed online.

9. Difference Between git clone and git remote add origin

git clone

Used when repository already exists online.

Example:

```
git clone https://github.com/user/project.git
```

Meaning:

```
GitHub → Local System
```

git remote add origin

Used when local project already exists.

Example:

```
git remote add origin URL
```

Meaning:

```
Local System → Connect to GitHub
```

10. git add

Command Used

```
git add .
```

Purpose

Tracks all files for commit.

Real-World Example

Suppose you created:

- login test
- API utility
- framework config

`git add .` prepares them for saving.

11. git commit

Command Used

```
git commit -m "Initial SDET journey setup"
```

Purpose

Creates snapshot/version of project.

Real-World Example

Think of commit as:

"Save point in game"

If future code breaks, older commit can restore project.

12. git push

Command Used

```
git push -u origin main
```

Purpose

Uploads local code to GitHub.

Real-World Example

Team members can now see latest framework updates.

13. Playwright Setup

Command Used

```
npm init playwright@latest
```

Purpose

Creates Playwright automation project.

What Gets Installed?

- Playwright package
- browsers
- config
- sample tests

Real-World Example

Used for:

- UI automation
 - regression testing
 - login validation
 - checkout flow automation
-

14. Why Playwright?

Advantages

- fast

- modern
- reliable
- async support
- multiple browser support

Real-World Example

Automating:

```
Amazon login  
Cart flow  
Checkout flow
```

15. Postman Setup

Purpose

Used for API testing.

What APIs Mean

Applications communicate using APIs.

Example:

```
Mobile App ↔ Backend Server
```

Real-World Example

Login API:

```
POST /login
```

returns:

```
{  
  "token": "abc123"  
}
```

Postman helps validate:

- status code
 - response body
 - authentication
-

16. Appium Setup

Command Used

```
npm install -g appium
```

Purpose

Installs Appium globally.

What is Appium?

Mobile automation framework.

Used for:

- Android automation
- iOS automation

Real-World Example

Automating:

- login
 - file upload
 - gestures
 - notifications
 - mobile flows
-

17. Android Studio Setup

Purpose

Provides:

- Android SDK
- adb
- emulator
- platform tools

Why Important?

Appium needs Android communication tools.

18. adb Setup

What is adb?

Android Debug Bridge.

Purpose

Allows laptop ↔ Android device communication.

Command Used

```
adb devices
```

Output

```
HGR5HJ6L    device
```

Meaning

Device connected successfully.

Real-World Example

Appium uses adb internally to:

- tap screen
 - install apps
 - open apps
 - capture logs
-

19. USB Debugging

Why Enabled?

Allows automation tools to control phone.

Real-World Example

Without USB debugging:

- Appium cannot automate mobile app.
-

20. Appium Server

Command Used

```
appium
```

Purpose

Starts Appium server.

Output

```
Appium REST http interface listener started
```

Real-World Example

Automation scripts send commands to Appium server.

Example:

```
click login button  
enter username  
scroll page
```

21. UiAutomator2 Driver

Command Used

```
appium driver install uiautomator2
```

Purpose

Allows Appium to automate Android devices.

Why Needed?

Appium server alone cannot automate devices. Driver acts as bridge.

Real-World Example

When automation clicks login button:

```
Script → Appium → UiAutomator2 → Android Device
```

22. Appium Inspector

Purpose

Inspect mobile app elements.

Used to find:

- id
- xpath
- accessibility id

- text

Real-World Example

Finding locator for:

Login button

Example:

```
await driver.$('id=loginBtn').click();
```

23. Issues Faced and Fixes

Issue 1 — npm Execution Policy Error

Problem

PowerShell blocked npm execution.

Fix

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Issue 2 — Git Command Confusion

Problem

Mixed:

```
git add
```

and:

```
git remote add origin
```

Fix

Used commands separately.

Issue 3 — LF and CRLF Warning

Problem

Windows line-ending warning.

Fix

Safe to ignore.

Issue 4 — adb not recognized

Problem

PATH variable not configured.

Fix

Verified adb manually using:

```
.\adb.exe version
```

Issue 5 — Empty Folders Not Showing in GitHub

Problem

Git does not track empty folders.

Fix

Need files like:

```
README.md
```

inside folders.

24. Real-World Workflow Going Forward

Daily Learning Plan

JavaScript

- async/await
 - coding problems
 - debugging
-

Playwright

- UI automation
 - waits
 - assertions
 - framework building
-

Appium

- Mira APK automation
 - gestures
 - mobile locators
 - waits
-

API Testing

- GET APIs
 - POST APIs
 - tokens
 - validations
-

GitHub

- branches
 - commits
 - framework enhancement
-

25. Branch Strategy

Future Branches

```
main
javascript-practice
coding-problems
playwright-learning
appium-learning
api-testing
framework-enhancement
```

Why Branches?

Allows experimentation safely.

Real-World Example

Framework enhancement can happen without breaking stable code.

26. Final Outcome Goal

Target

```
Manual QA → Strong SDET
```

Skills Being Built

- JavaScript
 - Playwright
 - Appium
 - API Testing
 - Git/GitHub
 - Automation Frameworks
 - Debugging
 - Interview readiness
-

27. Key Learning From Day 1

Strong setup matters.

Real SDETs:

- debug setup issues
- configure tools
- manage Git
- automate workflows
- continuously improve frameworks

Today's setup created the foundation for:

- real automation
- hands-on learning
- GitHub portfolio
- future interviews
- SDET growth path